# Prodigy Anesthesia Quick Review

## Thermoregulation

| | |
|---|---|
| What are the phases of heat loss and how are they characterized? | Core body temperature usually decreases 1-2 degrees Celsius during the first hour of anesthesia (Phase I), then gradually decreases over the next 3-4 hours (Phase II), then a consistent, steady-state or equilibrium temperature is reached (Phase III).<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1184.* |
| What temperature defines hypothermia? | Anything less than 36 degrees Celsius.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 321.* |
| Does regional anesthesia affect the thermoregulatory mechanism? | Yes, by blocking the thermoregulatory afferent and efferent nerve fibers, regional anesthesia interferes with sweating, vasoconstriction, and shivering responses.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1508.* |
| What nerve fibers mediate the sweat response? | Cholinergic sympathetic fibers<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1507.* |
| Why do patients describe a warming sensation on injection of neuraxial local anesthetics? | The blockade of the thermoregulatory afferent nerve fibers causes the hypothalamus to incorrectly interpret the peripheral temperature as high.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1508.* |
| What are four drugs that have been used successfully to prevent shivering? | Demerol, clonidine, ketanserin, and dexmedetomidine<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1509.* |
| What are the four routes of heat loss? Name them in the order from greatest degree of heat loss to least. | Radiation, convection, conduction, and evaporation<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |
| What temperature change stimulates the sweat response? | An increase in temperature of 0.9 degrees F (0.5 degrees C) produces a 7-fold increase in the sweat rate.<br>*Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1507.* |

*Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.*

# Prodigy Anesthesia Quick Review

## Thermoregulation

| | |
|---|---|
| Explain radiant heat loss. Where does it occur to the greatest degree in the human body? | Radiant heat loss is the transfer of electromagnetic radiation from the body to the cooler ambient environment. Blood carries heat from the body's core to the skin where it is emitted as infrared radiation. It occurs to the highest degree in the head because of the high percentage of blood flow it receives.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |
| How does radiant heat loss occur? | Radiant heat loss is the transfer of heat from the body into the cooler operating room.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |
| What is evaporative heat loss? | Evaporative heat loss occurs when moisture from the skin, exposed organs, and the respiratory tract turns from the liquid to gas phase. This process requires energy and the majority of the energy comes from our bodies. As a result of the loss of energy, our body loses heat as well.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |
| How does evaporative heat loss occur? | Evaporative heat loss occurs when liquids on the surface of the body evaporate into the air.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222-223.* |
| What is convective heat loss? | Convection is the generation of air currents by heat. As heat is emitted, it rises in a current and is replaced by cooler air.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |
| What is conductive heat loss? | Conduction is the transfer of heat energy from one object to another through physical contact. This is the method of heat loss that occurs when a patient is placed on a cold operating room table. Heat from the patient is lost to the colder table surface.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |
| How do convective and conductive heat loss occur? | Convective heat loss occurs as cooler air moves across the body's surface and conductive heat loss occurs when the body is in contact with a cooler surface such as an operating room table.<br>*Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 560.* |
| What type of heat loss accounts for most of the heat lost during surgery? | Radiant heat loss accounts for most of the heat lost in a surgical patient.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 222.* |

*Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.*

## Thermoregulation

| | |
|---|---|
| What temperature defines hyperthermia? | Anything exceeding 38 degrees Celsius.<br><br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 321.* |

*Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.*